SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 1 – Pseudocode Design

Course Code:	**DSA0404**	**Course Title:**	**FUNDAMENTALS OF DATA SCIENCE**
CO Assessed:	**CO5 –Pseudocode Design**	**Assessment:**	**Assessment Tool1 – Pseudocode Design**
Weightage:	**30% of CIA**	**Total Marks:**	**100**
CO5 Statement:	Design and develop pseudocode for implementing and evaluating basic Machine Learning techniques, including Linear Regression, Logistic Regression, and Unsupervised Learning. (BL4)		
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Section / Batch:	B.E-CSE-AI	**Date:**	24-08-2026

Code of Conduct

I ____________Shaik Nuha(192472242)_______ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: ______Sk.Nuha_____________

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning	Marks
Understanding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding; some issues missed	Poor understanding; problem not identified correctly	
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts	
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning	
Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided	
Clarity	10	Clear, well-structured	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand	

11. Linear Regression

Design pseudocode to train a linear regression model, calculate predicted values and residuals, and determine the Mean Squared Error (MSE) and R^2 score.

Problem Statement

Design a program to train a linear regression model, calculate predicted values and residuals, and determine Mean Squared Error (MSE) and R^2 score.

Algorithm

1. Start.
2. Read the input values X and target values Y.
3. Calculate the mean of X and Y.
4. Calculate the slope m.
5. Calculate the intercept c.
6. Predict values using:
   Y = mX + c
7. Calculate residuals:
   Residual = Actual - Predicted
8. Calculate MSE.
9. Calculate R^2 score.
10. Display the results.
11. Stop.

Code / Pseudocode

```
START
Input X = [1, 2, 3, 4, 5]
Input Y = [2, 4, 5, 4, 5]
Calculate meanX
Calculate meanY
Calculate:
m = Σ((X - meanX) * (Y - meanY)) /
   Σ((X - meanX)^2)
c = meanY - m * meanX
FOR each value in X
   predicted = m * X + c
```

residual = Y - predicted
  Display predicted and residual
END FOR
MSE = Σ((Y - predicted)^2) / n
R2 = 1 - [Σ((Y - predicted)^2) / Σ((Y - meanY)^2)]
Display m
Display c
Display MSE
Display R2
STOP

Output

Slope = 0.6

Intercept = 2.2

Predicted Values:

2.8

3.4

4.0

4.6

5.2

Residuals:

-0.8

0.6

1.0

-0.6

-0.2

MSE = 0.56

R^2 Score = 0.3077

12. Logistic Regression

Design pseudocode for logistic regression to predict whether a student will pass or fail based on study hours and attendance percentage.

Problem Statement

Design pseudocode for logistic regression to predict whether a student will pass or fail based on study hours and attendance percentage.

Algorithm

1. Start.
2. Input study hours and attendance.
3. Initialize logistic regression weights.
4. Calculate the linear combination.
5. Apply the sigmoid function.
6. If probability ≥ 0.5, predict Pass.
7. Otherwise, predict Fail.
8. Display probability and prediction.
9. Stop.

Code / Pseudocode

```
START
Input studyHours
Input attendance
Set w1 = 0.5
Set w2 = 0.05
Set bias = -5
z = (w1 * studyHours) + (w2 * attendance) + bias
probability = 1 / (1 + EXP(-z))
IF probability >= 0.5 THEN
   result = "PASS"
ELSE
   result = "FAIL"
END IF
Display probability
Display result
STOP
```

Example Input

Study Hours = 8

Attendance = 80%

Output

Probability = 0.9820

Prediction = PASS

13. Logistic Regression

Design pseudocode that calculates the sigmoid probability for logistic regression and converts the probability into a binary class using a threshold of 0.5.

Problem Statement

Design pseudocode to calculate the sigmoid probability for logistic regression and convert the probability into a binary class using a threshold of 0.5.

Algorithm

1. Start.
2. Read the value z.
3. Calculate sigmoid:
   P = 1 / (1 + e^-z)
4. Compare probability with 0.5.
5. If probability ≥ 0.5, class = 1.
6. Otherwise, class = 0.
7. Display probability and class.
8. Stop.

Code / Pseudocode

```
START
Input z
probability = 1 / (1 + EXP(-z))
IF probability >= 0.5 THEN
   class = 1
ELSE
   class = 0
END IF
```

Display probability

Display class

STOP

Input:

z = 2

Output

Sigmoid Probability = 0.8808

Binary Class = 1

For example:

Probability >= 0.5 → Class 1

Probability < 0.5 → Class 0

14. Logistic Regression

Design pseudocode to evaluate a logistic regression classifier using the confusion matrix, accuracy, precision, recall, and F1-score.

Problem Statement

Design pseudocode to evaluate a logistic regression classifier using a confusion matrix, accuracy, precision, recall, and F1-score.

Algorithm

1. Start.
2. Input actual and predicted class values.
3. Calculate:
    - True Positive (TP)
    - True Negative (TN)
    - False Positive (FP)
    - False Negative (FN)
4. Calculate accuracy.
5. Calculate precision.
6. Calculate recall.
7. Calculate F1-score.
8. Display all metrics.
9. Stop.

Code / Pseudocode

```
START
Actual    = [1, 1, 1, 0, 0, 0, 1, 0]
Predicted = [1, 1, 0, 0, 0, 1, 1, 0]
TP = 0
TN = 0
FP = 0
FN = 0
FOR each value
   IF Actual = 1 AND Predicted = 1
      TP = TP + 1
   ELSE IF Actual = 0 AND Predicted = 0
      TN = TN + 1
   ELSE IF Actual = 0 AND Predicted = 1
      FP = FP + 1
   ELSE IF Actual = 1 AND Predicted = 0
      FN = FN + 1
   END IF
END FOR
Accuracy = (TP + TN) / (TP + TN + FP + FN)
Precision = TP / (TP + FP)
Recall = TP / (TP + FN)
F1 = 2 * Precision * Recall /
    (Precision + Recall)
Display confusion matrix
Display Accuracy
Display Precision
Display Recall
Display F1-score
STOP
```

Output

For the given values:

Confusion Matrix:

		Predicted	
		1	0
Actual	1	3	1
	0	1	3

TP = 3

TN = 3

FP = 1

FN = 1

Accuracy = 0.75

Precision = 0.75

Recall = 0.75

F1 Score = 0.75

15. Unsupervised Learning

Design pseudocode for an unsupervised learning process that accepts an unlabelled dataset, preprocesses the features, discovers groups, and produces cluster assignments.

Problem Statement

Design pseudocode for an unsupervised learning process that accepts an unlabelled dataset, preprocesses the features, discovers groups, and produces cluster assignments.

Algorithm

1. Start.
2. Load the unlabelled dataset.
3. Select relevant features.
4. Handle missing values.
5. Normalize/standardize the features.
6. Choose the number of clusters K.
7. Initialize cluster centroids.
8. Assign each data point to the nearest centroid.
9. Recalculate centroids.
10. Repeat assignment and centroid calculation until convergence.

11. Produce cluster assignments.
12. Display the clusters.
13. Stop.

Code / Pseudocode

```
START
Input unlabelled dataset
Dataset =
[10, 20]
[12, 22]
[11, 19]
[80, 90]
[82, 88]
[78, 92]
Preprocess the data
Handle missing values
Normalize features
Set K = 2
Initialize two cluster centroids
REPEAT
   FOR each data point
      Calculate distance from each centroid
      Assign data point to nearest centroid
   END FOR
   Recalculate centroid of each cluster
UNTIL centroids do not change
Display cluster assignments
STOP
```

Output

Number of Clusters = 2

Cluster 1:

(10,20)

(12,22)

(11,19)

Cluster 2:

(80,90)

(82,88)

(78,92)